MILITARY SPECIFICATION

RECEIVING SET, RADIO - AN/WRR-3( )

1. SCOPE

1.1 The radio receiving set covered by this specification shall provide for the reception of radio frequency signals in the frequency range of 14 kilohertz (kHz) to 600 kHz. It shall be of such design and construction as to be suitable for installation aboard U.S. Navy surface ships and submarines of all types and classes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901	–	Shock Tests, H. I. (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-3098	–	Crystal Units, Quartz, General Specification for
MIL-Q-9858	–	Quality Program Requirements
MIL-M-10304	–	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-E-16400	–	Electronic Equipment, Naval Ship and Shore: General Specification
MIL-F-19207	–	Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification for

STANDARDS

MILITARY

MIL-STD-108	–	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-189	–	Racks, Electrical Equipment, 19-inch and Associated Panels
MIL-STD-200	–	Electron Tubes, Selection and Use of
MIL-STD-242	–	Electronic Equipment Parts (Selected Standards)
MIL-STD-243	–	Type and Definitions of Models for Communications Electronic Equipments
MIL-STD-275	–	Printed Wiring for Electronic Equipment
MIL-STD-415	–	Test Points and Test Facilities for Electronic Systems and Associated Equipment Standard for
MIL-STD-454	–	Standard General Requirements for Electronic Equipment
MIL-STD-461	–	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-781	–	Reliability Tests: Exponential Distribution
MIL-STD-785	–	Requirements for Reliability Program (for Systems and Equipments)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

MILITARY

MIL-HDBK-217 — Reliability Stress and Failure Rate Data for Electronic Equipment

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 First article sample. - Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.2 General requirements. - The equipment shall meet the requirements of MIL-E-16400, except as hereinafter specified.

3.2.1 Material. - All material shall be in accordance with MIL-E-16400, except as may otherwise be specified herein. Particular attention shall be paid to the acceptable and unacceptable materials of MIL-E-16400.

3.2.2 Selection of parts. - The selection and application of all parts shall be in accordance with MIL-E-16400.

3.2.2.1 Nonstandard parts. - The preparation and submission of data necessary for the evaluation of nonstandard electronic parts shall be in accordance with MIL-E-16400. Approval of nonstandard parts, previously granted on prior procurements of the AN/WRR-3( ) equipments, or for nonstandard parts employed in the Government-furnished equipment, shall not be construed as being applicable to equipment covered by this specification, nor as waivers of any requirement of the latter.

3.2.3 Temperature. - The radio set shall conform with the requirements of MIL-E-16400 for class 4 temperature.

3.2.4 Test points. - Test points shall be provided, in accordance with MIL-E-16400, to the extent where compliance with these requirements are compatible with the test points provided in the Government-furnished equipment.

3.2.5 Electron tubes, semiconductor devices (diode). - All electron tubes shall be selected from among the approved types of 6.3 volt miniature types, in accordance with MIL-E-16400. All electron tube shields used on miniature electron tubes shall be of the heat dissipating types and shall be of nonmagnetic materials.

3.2.5.1 Semiconductor devices (diode). - Semiconductor devices (diode) shall be selected from the approved types, in accordance with MIL-E-16400.

3.2.5.2 The Government-furnished equipment shall dictate the selection and application of all electron tubes and semiconductor devices (diode). Preferred electron tubes and semiconductor devices, where employed in the Government-furnished equipment, shall be duplicated in equipments covered herein, with respect to individual types and circuit applications. The use of nonpreferred types, irrespective of their use in the Government-furnished equipment, shall be specifically approved in accordance with MIL-E-16400.

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3.2.6 Fuses. - Each side of the input to the primary power supply shall be separately fused. The fuses shall be of the miniature glass tube type, and shall be mounted in insulated fuse holders on the front operating panel. Fuses shall be replaceable from the front of the equipment without the necessity for opening the equipment.

3.2.7 Blown fuse indicators. - Blown fuse indicators, conforming to MIL-F-19207, shall be provided for all active fuses. A spare fuse holder and blown fuse indicator shall be provided on the front operating panel. The spare blown fuse holder shall be covered with a readily removable black masking coating to distinguish it from those associated with the active fuses.

3.2.8 Time meters. - Elapsed time meters will not be required.

3.2.9 Control knobs. - Tactile identification forms of control knobs will not be required. Control knobs, other than those for the main tuning control and frequency band switch, shall conform with MIL-STD-242. The control knobs for the main tuning and frequency band switch controls shall duplicate their counterparts on the Government-furnished equipment. The main tuning control shall be provided with a locking device of such design as to afford positive locking and unlocking actions, without the aid of tools and without maladjustment of the tuning control adjustment.

3.2.10 Enclosure. - The degree of enclosure shall be totally enclosed, in accordance with MIL-STD-108.

3.2.11 External finish. - External surfaces shall be finished in accordance with MIL-E-16400.

3.2.12 Printed wiring. - Printed wiring shall be employed in the same manner as for the Government-furnished equipment. The fabrication of printed circuit boards, and the insertion of parts and assemblies thereon, shall conform to the requirements of MIL-E-16400, except that the dimensional requirements of MIL-STD-275 are not applicable in those instances where such application would involve redesign of the counterpart printed circuit boards of the Government-furnished equipment.

3.2.13 Soldering. - All soldering of wiring to terminal connectors shall conform to MIL-E-16400. Insulating sleeves (spaghetti) shall be installed on all pins and jacks and all other closely spaced solder type terminations.

3.2.14 Securing devices. - All screws and nuts, employed in the fabrication of the receiving set, shall be of austenitic corrosion-resistant steel.

3.2.14.1 All machine screws and bolts shall employ Phillips type heads. Captive type screws or bolts shall be employed for securing all plug-in electronic assemblies, shield plates or dust covers and shall be captive with the removable assembly or part.

3.2.14.2 Cadmium-plated screws, bolts, nuts, and other threaded hardware devices shall not be used, MIL-E-16400 notwithstanding.

3.2.15 Crystals. - All crystals employed in the equipment shall conform to MIL-E-16400.

3.2.16 Electrical indicating meters. - Front panel counted electrical indicating meters shall, in general, conform to the requirements of MIL-E-16400. The dials shall employ white markings on black backgrounds. Radioactive luminescent markings shall not be employed. The tuning meter shall conform to MIL-M-10304/24. The a.f. output decibel meter shall conform to MIL-M-10304/19.

3.2.17 Panel illumination. - Panel illumination, other than that required for the digital frequency indicators, associated with the main tuning and frequency band switch controls, is not required. A dimmer control is not required. The illumination of the digital frequency indicators, the arrangement and mounting of the

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dial lamps (including spares) and the accessibility of the dial lamps for replacement shall duplicate their counterparts in the Government-furnished equipment.

3.2.18 Resilient mountings. - The radio receiving set shall be capable of operation and performance in accordance with the requirements of this specification when "hard mounted" on top of an operating table without the use of resilient mountings of any description; i.e., shock or vibration mounts shall not be employed.

3.2.19 Iron cored transformers and inductors. - All audio and power transformers and inductors shall conform to the requirements of MIL-E-16400, as applicable to shipboard electronic equipment.

3.2.19.1 The audio output transformer shall conform to the following:

(a) The primary winding shall be electrostatically shielded from the secondary windings.
(b) The secondary winding, feeding the head telephone jack, shall have one end of the winding at ground potential.
(c) The secondary winding, feeding the rear mounted line output receptacles, shall be electromagnetically balanced, with respect to ground. The center tap of this winding shall terminate at a separate solder type terminal on the transformer assembly. This terminal shall afford optional connection to ground, by soldering methods. It shall be left ungrounded in the equipments, as delivered.

3.2.20 RF filter. - Radio frequency (r.f.) filters shall be provided between the power input connector and the input to the power transformer; between the r.f. output connector and the "final" intermediate frequency (i.f.) amplifier input; and between the a.f. line output connectors and the audio frequency (a.f.) output transformers. The filters providing r.f. filtering of the power supply input and r.f. line outputs shall be balanced with respect to ground.

3.3 Composition. - The equipment shall contain all parts necessary for its performance in accordance with the requirements of this specification when connected to the specified primary power source.

3.3.1 Cable plugs and applicable cable clamps to mate with external connectors, except headphones plug, shall be furnished as a part of each equipment.

3.3.2 One test cable assembly, to complete all electrical circuits between the chassis assembly and the rear cabinet mounted connectors, when the chassis assembly is withdrawn for servicing, shall be provided with each equipment.

3.3.3 One test prod to facilitate connection to test points shall be furnished.

3.3.4 Head telephones and plugs shall not be furnished.

3.3.5 External cables for connection to the power source or to ancillary equipment shall not be furnished.

3.4 Overall dimensions and weight. -

3.4.1 Dimensions. - The maximum overall dimensions exclusive of front panel projections (handles, controls, and other devices), shall not exceed the following:

(a) Width - 17¼ inches
(b) Height - 8-3/4 inches
(c) Depth - 16-3/4 inches

3.4.2 Weight. - Weight shall not exceed 70 pounds, maximum.

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3.5 Power source. - The equipment shall be designed to operate from a primary power source of 105, 115, 125 volts, 50 to 60 Hz and 400 Hz, single phase, alternating current (ac), with voltage variations of ±10 percent with respect to the nominal voltage applied, and power line frequency variations of ±5 percent, with respect to the nominal power frequency.

3.5.1 The total power consumption at 115 volts, 60 or 400 Hz shall not exceed 60 watts.

3.5.2 The power supply shall employ solid state devices for rectification and voltage or current regulation. It shall perform the same functions as for its counterpart in the Government-furnished equipment.

3.5.3 The primary winding of the power transformer shall be electrostatically shielded. Solder type terminals shall be provided on the power transformer for each of the nominal power source voltages. Connection to the proper primary terminal shall be such that they may be readily accomplished, as an installation matter, by soldering methods.

3.6 Frequency range. - The nominal overall operating frequency range shall be from 14 to 600 kHz, without frequency hiatus.

3.6.1 Frequency bands. - The specified complete overall nominal operating frequency range shall be covered by five frequency bands, having nominal operating frequency ranges as specified in table I. Selection of frequency bands shall be accomplished by band switching methods.

TABLE I. Nominal operating frequency ranges of frequency band

Band No.	Nominal operating frequency range kHz
I	14 to 30
II	30 to 63
III	63 to 133
IV	133 to 283
V	283 to 600

3.7 Antenna input. - The radio frequency input circuits shall provide for external connection on a 50-ohm nominal impedance or a 200-micromicrofarad nominal capacitance impedance.

3.7.1 The antenna input connector and the facilities for transfer from low to high impedance r.f. input shall be the same as employed in the Government-furnished equipment.

3.8 Modes of reception. - The radio set shall provide for the following modes of reception:

A-1 Continuous wave telegraphy (CW)
A-2 Modulated continuous wave telegraphy from 250 to 600 kHz only
F-1 Frequency shift keying (FSK), using external converter equipment

3.9 Operational requirements. - The radio receiving set and all parts or items entering into its construction shall be designed for continuous operation, in conformity with the performance requirements of the specification, under the adverse conditions normally encountered aboard U. S. Navy surface ships and submarines of all types and classes. Continuous operation shall be interpreted as meaning full operation, without power turn off, except for maintenance or overhaul.

3.10 Electrical circuitry. - The complete circuitry of the radio set, including all facilities provided for alignment or like adjustments, shall be of the same configuration and shall employ the same principles of design and operation as for the Government-furnished AN/WRR-3( ) equipment. Deviations from these requirements, without prior approval by the command or agency concerned will not be acceptable.

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3.11 Interchangeability. - Except as otherwise noted, the design and construction of the radio receiving set and all subassemblies and parts shall afford electrical and mechanical interchangeability (as pertinent) with their counterpart subassemblies and parts of the Government-furnished equipment to the extent that such interchangeability is compatible with the requirements of MIL-E-16400 and the requirements of this specification for performance, additionally:

3.11.1 In all instances, the performance requirements of this specification shall take precedence and shall govern in cases of conflicts or inconsistency between the specified performance requirements and corresponding performance characteristics demonstrated by the Government-furnished equipment.

3.11.2 Nonstandard parts, present in the Government-furnished equipment, shall not be employed in those applications where electrically and mechanically interchangeable standard parts are currently available.

3.11.3 Those electrical and mechanical designs of the r.f. and first conversion oscillator coils in the r.f. and first conversion oscillator subassemblies shall duplicate their counterparts in the Government-furnished equipment.

3.12 Chassis latching mechanism. - The latching mechanisms, provided on the front operating panel to secure the chassis within the cabinet, shall be of such design as to preclude unlatching when the equipment is subjected to shock or vibration, as specified herein. Although the operational and latching features of these mechanisms shall be the same as for those employed on the Government-furnished equipment, they need not require the same mounting dimensions.

3.13 Relay rack mounting brackets. - The cabinet assembly shall contain "blind" tapped inserts for receiving the securing screws for mounting brackets to permit the mounting of the equipment in relay racks or cabinets conforming to MIL-STD-189.

3.13.1 Relay rack mounting brackets shall not be furnished. However, the contractor shall prepare instructions and drawings, for inclusion in the technical manual, to permit the fabrication of such brackets by a commercial or Navy shipyard. The drawings and instructions shall be submitted for approval by the command or agency concerned prior to the fabrication of the first article model.

3.14 Special tools.- Securing devices which require special tools for their tightening, loosening, or any other forms of manipulation shall not be employed. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.15 Reliability. -

3.15.1 Quantitative reliability requirement. - The minimum acceptable MTBF (as defined by MIL-STD-781) of this equipment shall be 300 hours.

3.15.2 Reliability program plan. - The manufacturer shall submit a reliability program plan in accordance with and in paragraph sequence with MIL-STD-785, except for those paragraphs pertaining to "program review and development testing", which are not applicable. Particular attention shall be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks, and time schedules. The proposed reliability program plan shall be submitted as a separate and complete entity within the total equipment delineation. The reliability program plan, as approved, becomes the basis for contractual compliance. The plan, including reliability test plans, must describe an integrated effort within the total project; it shall provide information to show how the manufacturer will meet specified quantitative reliability requirements during manufacture.

3.15.3 Reliability prediction. - The manufacturer shall submit a reliability prediction in accordance with paragraph titled "How to make a reliability prediction based on stress factors and part populations" of MIL-HDBK-217 to the command or agency concerned. The prediction shall assume a 65° C temperature and

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a shipboard environment. The contractor shall evaluate his prediction with respect to the reliability he expects from a production unit. If there is a discrepancy between the predicted MTBF and specified MTBF, the contractor shall identify the causes of the discrepancies and instigate action to assure that the inherent reliability of the equipment is preserved during the production process. The processes and procedures required to assure production of reliable equipment shall be thoroughly documented.

3.15.4 Source of failure rates. - The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.15.5 Reliability reports. - Reports, providing a complete accounting of progress on each element, defined by the reliability program plan, results achieved, and status of actions to resolve problems and correct weak links, shall be submitted monthly to the command or agency concerned. The results of the demonstration test shall be reported in accordance with paragraph titled "Reliability Test Reports" of MIL-STD-781.

3.16 Performance requirements. - The radio receiving set shall conform to the following requirements for performance, when the set is operated under the specified standard test conditions, unless otherwise indicated. (See 4.5 thru 4.5.3.10.)

3.16.1 Frequency overlap. - The total frequency overlap between adjacent frequency bands shall not be less than 2.5 percent of their common nominal end frequencies, or less than 1.0 percent between the same nominal end frequency and the highest frequency covered by the lower of the two adjacent frequency bands, or the lowest frequency covered by the higher of these two frequency bands. The frequency overlap at the nominal low and high frequencies of the complete frequency range of the receiver shall not be less than 1.0 percent of the nominal end frequencies.

3.16.1.1 The limits for frequency overlap between adjacent frequency bands shall be as specified in table II.

TABLE II. Total frequency overlap between adjacent frequency bands

Common nominal end frequency kHz	Minimum total frequency overlap kHz
30	0.750
63	1.575
133	3.325
283	7.075

3.16.1.2 The limits for frequency overlap at the ends of each frequency band shall be as specified in table III.

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TABLE III. Frequency overlap at ends of each frequency band

Nominal end frequency kHz	Minimum frequency overlap kHz
14	0.14
30	0.30
63	0.63
133	1.33
283	2.83
600	6.00

3.16.2 Accuracy of digital main tuning indicator calibration. - The accuracy of the digital main tuning indicator calibration shall be within the limits, specified in table IV, of the actual resonant frequency to which the receiving set is tuned. The specified limits for indicator accuracy shall be applicable to any frequency adjustment over the nominal frequency range of the pertinent frequency band. The specified limits shall be interpreted as including all errors contributed by the mechanical backlash of the main tuning drive assembly. The specified limits are predicated upon the preadjustment of the indicator calibration to the calibration check point, nearest to the resonant frequency adjustment of the receiver, by means of the self-contained calibrator of the radio receiving set. The specified limits are applicable to clockwise or counter-clockwise rotation of the main tuning control with the receiver conditioned as for A-1 Broad mode of reception and A-1 sensitivity. The allowable error is the difference between the frequency of the applied signal and the indicator reading for that signal.

TABLE IV. Accuracy of main tuning indicator calibration

Frequency band	Nominal frequency range kHz	Allowable frequency error Hz
I	14 to 30	±100
II	30 to 63	±100
III*	63 to 133	±100
IV*	133 to 283	±150
V *	283 to 600	±250

*Note: Over the lowest $2\frac{1}{2}$ per cent of the nominal frequency range of bands III, IV, and V the accuracy of frequency indication shall be within -115 percent or 350 Hz, whichever is the least exacting.

3.16.2.1 Manipulation of the mechanical dial lock for the main tuning control, to either its locked or unlocked conditions, shall not abrogate the limits specified in table IV.

3.16.3 Linearity of tuning variation. - Using the average frequency variation for 0.1 percent of the complete nominal frequency range of any frequency band as a reference, the frequency change for any 10 percent angular rotation of the ganged tuning capacitors shall be within ±10 percent of the reference value. Compliance with this performance requirement shall be determined upon the use of an arbritrary linear scale, attached to the main tuning capacitor drive shaft and with frequency measurements made at every 10 percent of angular rotation of the ganged tuning capacitors, beginning and ending at the lowest and highest nominal frequency, respectively, of the frequency band under examination.

3.16.4 Internal calibrator. - The front panel control and the operation of the internal calibrator shall be such as to provide calibration check points every 10 kHz over the complete nominal frequency range of the receiver. In addition, calibrations shall be effective without any evidence of ambiguity.

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3.16.5 Sensitivity. - The sensitivity shall not be poorer than the values specified in table V, when measured in accordance with 4.5.

TABLE V. Sensitivity microvolts

Band	High impedance input mode of reception		F-1	A-2*
	A-1 Broad	A-1 Sharp		
I	5.0	4.0	5.0	X
II	4.0	3.0	3.0	X
III	4.0	3.0	3.0	X
IV below 250 kHz	4.0	3.0	3.0	X
IV above 250 kHz	3.0	3.0	3.0	X
V	3.0	2.0	2.0	3.0
I through V	Low impedance input			
	1.0	0.5	0.5	1.5

*For A-2, the specified limits shall be applicable only to the frequency range from 250 to 600 kHz.

3.16.6 Gain variation. - The difference between maximum overall gain and minimum overall gain, for any mode of reception, when the receiver is tuned from one nominal end frequency to the opposite nominal end frequency of any frequency band, shall not exceed 10 decibels (db), with a constant level radio frequency input signal applied to either the high impedance or low impedance antenna input, and with all controls, except the main tuning control, remaining in a fixed preset adjustment.

3.16.7 Reserve gain. - The overall gain of the equipment, with the r.f. gain control set for maximum gain, shall be at least 20 db greater than the gain obtained with the r.f. gain control adjusted for sensitivity measurement, at the frequency of minimum gain, in each frequency band, as determined from the gain variation measurement.

3.16.8 Preselection selectivity. - The selectivity offered by all of the r.f. tuned circuits between the antenna input connector (for either high or low impedance input) and the first converter shall fall within the limits shown in table VI.

TABLE VI. Preselection selectivity (maximum bandwidth)

Frequency kHz	kHz Off resonance - above and below			
	6 db down	20 db down	40 db down	60 db down
14	0.21	0.60	1.25	3.50
30	0.36	0.90	2.10	5.70
63	0.62	1.60	4.20	10.00
133	1.06	3.20	8.25	20.00
283	2.26	6.30	16.40	40.00
600	4.80	12.50	32.00	75.00

3.16.9 Overall r.f. and i.f. selectivity. - The selectivity offered by all tuned circuits and filters included between the antenna connector and the input to the final detector shall fall within the limits depicted in table VII, with the i.f. selectivity control at broad and the receiver conditioned for A-1 Broad reception. This requirement shall apply to either high or low impedance antenna connections.

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TABLE VII. Overall selectivity (maximum bandwidth)

Frequency kHz	kHz Off resonance - above and below			
	6 db down	20 db down	40 db down	60 db down
14	0.20	0.50	1.00	1.60
30	0.35	0.95	1.70	2.30
63	0.62	1.35	2.15	3.00
133	1.00	1.85	2.50	3.40
283	1.50	2.30	3.20	4.30
600	1.80	2.70	3.70	4.80

3.16.9.1 Minimum bandwidth. - The minimum bandwidths shall be consistent with the modulation requirements.

3.16.10 Final i.f. amplifier selectivity. - The final i.f. amplifier shall offer 2 degrees of i.f. selectivity, obtained through the use of mechanical filters preceding a wideband amplifier. The selectivity obtained from the mechanical filters shall conform with the following limits for the broad and sharp positions of the i.f. selectivity control.

(a) Broad i.f. selectivity - Total bandwidth at 6 db down - 3.0 ±0.5 kHz. The ratio of the total bandwidth at the 60 db point of attenuation to the total bandwidth at the 6 db point of attenuation shall not exceed 2.3 to 1.

(b) Sharp i.f. selectivity - Total bandwidth at 6 db down - 1.0 ±0.2 kHz. The ratio of the total bandwidth at the 60 db point of attenuation to the total bandwidth at the 6 db point of attenuation shall not exceed 3.0 to 1.

3.16.10.1 Undulations of transmission loss across the pass band of either the broad or sharp filters shall not exceed 3 db.

3.16.10.2 The center of resonance of the broad and sharp i.f. selectivities, as determined from measurements of frequencies at the 6 db points of attenuation, shall not differ from the nominal intermediate center frequency by more than 10 percent of the actual measured 6 db bandwidth (total) when the i.f. selectivity control is changed from broad to sharp and vice versa.

3.16.10.3 The output noise level, as based upon a change in overall i.f. gain, when switching between broad and sharp setting of the i.f. selectivity control, shall not change by more than 6 db.

3.16.11 Overload (desensitization) selectivity. - The frequency separation between the frequencies of a weak desired signal and a strong interfering signal, both above and below resonance, shall not exceed the limits shown in table VIII when measured in accordance with 4.5.3.1.

TABLE VIII. Overload selectivity (frequency separation limits)

Resonant frequency	Interfering signal level, with respect to desired signal		
	60 db	80 db	100 db
kHz	kHz	kHz	kHz
14	1.6	2.2	2.9
30	2.6	3.6	4.7
63	4.2	5.8	7.6
133	6.7	9.6	12.3
283	11.1	15.9	20.4
600	16.5	25.8	32.4

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3.16.12 Cross modulation selectivity. - The frequency separation between the frequencies of a weak desired signal and a strong interfering signal, both above and below the resonant frequency, shall not exceed the limits shown in table IX when measured in accordance with 4.5.3.2.

TABLE IX. Cross modulation selectivity (frequency separation limits)

Resonant frequency	Interfering signal level, with respect to desired signal level		
	60 db	80 db	100 db
kHz	kHz	kHz	kHz
250	10	52.5	100
600	25	70.0	125

3.16.13 I.F. rejection. - The equipment shall offer not less than 70 db of attenuation to signals of the intermediate frequency over the nominal signal frequency range of 14 to 30 kHz, and not less than 85 db over the nominal signal frequency range above 30 kHz.

3.16.14 Image attenuation. - Primary and secondary image attenuation shall be not less than 100 db over the complete frequency range of the equipment.

3.16.15 Other spurious responses. - With no r.f. input, there shall be no spurious audio responses in excess of that which would result from a resonant r.f. input, applied to the low impedance input circuit, of 0.1 microvolt, except as follows.

3.16.15.1 The spurious outputs, resulting when the radio receiving set is tuned to the fundamental or harmonic frequency of any conversion oscillator, shall not exceed the equivalent of 10 percent of the specified sensitivity input at any frequency throughout the nominal frequency range of the equipment. Harmonics higher than the tenth harmonic need not be considered.

3.16.15.2 Intermodulation products, as defined in 4.5.2.9.4, shall not be less than 60 db down with respect to any desired resonant frequency over the complete nominal frequency range of the equipment.

3.16.16 Audio output voltage regulation. - The audio amplifier shall be so designed that the output voltage variation shall remain within ±30 percent (in voltage) of the value obtained with a 600-ohm noninductive resistance load, for any change in the resistive value of this load from 120 to 3000 ohms.

3.16.17 Resonant overload output. - Resonant overload output, (that is, the output where the overall gain of the receiver, as adjusted for standard A-1 or F-1 sensitivity, falls 3 db with increase of r.f. input level), shall be not less than 60 milliwatts of power into a 600-ohm line output load.

3.16.17.1 Under like conditions, but with the phone level control set at maximum output, the resonant overload output appearing across a 600-ohm noninductive resistive load, connected to the head phone jack, shall fall between 5 and 15 milliwatts.

3.16.17.2 The output level shall not fall below the actual measured resonant overload for any variation of continuous wave (c.w.) input up to 2.0 volts and for any variation of r.f. gain between the gain setting for sensitivity measurements and 60 db below such gain setting.

3.16.17.3 Under the conditions specified in 3.16.17, the overall gain shall not vary more than ±2 db for output levels from 100 microwatts up to 3 db below actual resonant overload output.

3.16.18 Maximum undistorted output. - Maximum undistorted output (7 percent distortion), as predicated upon adjustment of the radio receiver for A-2 mode of reception, shall be not less than 60 milliwatts into a 600-ohm line output load.

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3.16.18.1 With the phone level control set at maximum output, the maximum undistorted output (7 percent distortion) appearing across a 600-ohm noninductive resistance load connected to the head telephone jack, shall fall between 5 and 15 milliwatts.

3.16.18.2 The output level shall not fall below the actual measured, maximum undistorted output (7 percent distortion) for any variation of modulated continuous wave (m.c.w.) input up to at least 100 times the input required to reach actual maximum undistorted output.

3.16.19 Audio amplifier frequency response. - The overall a.f. response of the a.f. amplifier, for the broad and sharp conditions of a.f. selectivity, shall conform with the following requirements, using as the zero decibel point of reference level, 6 milliwatts of output power in a 600-ohm noninductive resistance load across the line output terminals and an output frequency of 1000 Hz.

(a) Broad a.f. selectivity

(1) From 250 to 3000 Hz	Flat within 6 db
(2) At 100 Hz	Down not less than 10 db
(3) At 50 Hz	Down not less than 35 db
(4) At 10,000 Hz	Down not less than 20 db

(b) Sharp a.f. selectivity

(1) Center of passband shall be 1000 Hz, ±25 Hz.
(2) Total bandwidth 6 db down shall be not less than 350 Hz.
(3) Total bandwidth at 40 db down shall be not more than 700 Hz.
(4) Attenuation of frequencies below 600 Hz shall be 40 db or more.
(5) Attenuation of frequencies above 1500 Hz shall be 40 db or more.

3.16.19.1 The insertion loss of the switchable a.f. bandpass filter shall be compensated so that switching from broad to sharp shall not cause a change of audio output in excess of 1 db.

3.16.19.2 The foregoing requirements for a.f. frequency response shall also apply for any output level from the lowest discernible output up to resonant overload output, and for any output load resistance within the limits specified in 3.3.16.

3.16.20 C. W. detector distortion. - The distortion produced by the c.w. detector shall be low enough that with the beat frequency oscillator (b.f.o.) adjusted for a 500 Hz beat note, and sharp a.f. selectivity, the audio output level into a 600-ohm external load will be down 30 db, or more, with respect to the audio output level, when the b.f.o. is readjusted to produce a 1000 Hz beat note.

3.16.21 Output signal-to-hum ratio. - Output signal to hum ratio for any mode of reception shall be not less than 40 db for any adjustment of the output level from maximum output down to where the output hum is 0.3 microwatt. This requirement shall be applicable to any specified primary power frequency.

3.16.21.1 The requirements of 3.16.21 shall apply to all hum and noise arising in the final detector and audio amplifier circuits, as well as the hum arising in other circuits, which produce hum modulation or similar effects on the r.f. signal.

3.16.22 Manual r.f. gain control. - The manual r.f. gain control shall have such a range as to permit a variation in the overall gain of the r.f. and i.f. amplifiers of not less than 100 db when the equipment is conditioned for the reception of A-1 and F-1 signals. It shall permit a continuous variation in output for any constant value of r.f. input. The specified range of gain variation shall be obtained with not less than 90 percent of a total of not less than 270 degrees of rotation of the control shaft.

3.16.22.1 Gain control linearity. - The variation of gain with respect to angular rotation of the control shaft shall be such that a curve plotted on linear graph paper of gain with respect to degrees of angular shaft rotation will, at all points, be within ±6 db of a straight line drawn between the point on the curve representing 100 db of gain reduction and the point on the curve representing the zero point of attenuation.

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3.16.22.2 The gain control circuits shall operate in such a manner that the overall selectivity shall not fall below the requirements of this specification when measured for any resonant input up to 100,000 microvolts, with the receiver gain reduced so as to maintain 6 milliwatts of output. This requirement does not prohibit a displacement of the center of resonance, provided that such displacement does not exceed 25 Hz.

3.16.23 Output limiter characteristics. - The output limiter shall operate to hold the audio output level constant, within a total tolerance of 2 db for any variation of c.w. input, at the antenna input connector, between the input required at the threshold of operation of the output limiter and 100,000 microvolts. This requirement shall be applicable for the following conditions:

(a) Any r.f. within the nominal range of the equipment
(b) Any output level within the range of the output limiter control
(c) Manual r.f. gain control adjusted as for sensitivity measurement
(d) Any audio beat note frequency within the range of the b.f.o. frequency vernier (see 3.16.28.2.1)

3.16.23.1 The overall distortion, including all factors such as amplifier or detector tube limiting and output limiter actions, shall not exceed 50 percent.

3.16.23.2 The output limiter shall be sufficiently rapid as to result in the effective limiting of each individual cycle of an output signal of any audio frequency up to 1500 Hz.

3.16.24 Output limiter level control range. - The output limiter control shall have such a range as to permit a variation of audio output level, as measured at the threshold of operation of the output limiter of not less than 40 db. This range shall be obtained in not less than 90 percent of the total of not less than 270 degrees of angular rotation of the control shaft.

3.16.24.1 Output limiter level control linearity. - The variation of audio output level with respect to angular rotation of the control shaft shall be such that a curve plotted on linear graph paper of audio output, with respect to degrees of angular rotation, shall be within ±6 db of a straight line drawn between the point on the curve representing 40 db reduction and the point on the curve representing 2 db reduction from the maximum audio output level.

3.16.25 Noise limiter. - The noise limiter shall effectively limit or reduce voltage surges, or peaks exceeding the output obtained with a 60 percent modulated signal, with no degradation of overall distortion, as specified herein.

3.16.26 Range and linearity of head telephone level control. - The a.f. output level control, associated with the head telephones, shall provide a linear relationship, within ±12 db between its angular rotation and a.f. output. It shall provide an attenuation of not less than 40 db over not less than 90 percent of a total of not less than 270 degrees of angular rotation. The linearity shall be determined using a linear coordinate graph of a.f. output level (in db) versus angular rotation of the control shaft. The control characteristics shall be compared to a straight line drawn between the 40 db point of attenuation and the point at which the characteristic curve departs from the point of zero attenuation.

3.16.27 Range and linearity of a.f. output level control. - The range and linearity of the output level control, employed for adjustment of a.f. line output level, when the output limiter is inoperative, shall be as specified in 3.16.24 and 3.16.24.1.

3.16.28 Beat frequency oscillator and frequency vernier. - The beat frequency oscillator shall be aligned for A-1 and F-1 modes of reception when the applied final i.f. signal is centered within ±0.01 percent of the midband frequency of the "final" i.f. amplifier, with the frequency vernier control set at "zero" and with a zero beat audio output.

3.16.28.1 The frequency of the b.f.o. with respect to the frequency of the final i.f. amplifier shall be such that when properly tuned for receiving an F-1 signal, a decrease in the frequency of the received signal will result in an increase in the frequency of the audio output beat note.

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3.16.28.2 The b.f.o. frequency vernier control shall cause an increase in the frequency of the b.f.o. when it is rotated in a clockwise direction with respect to its "zero" marking and a decrease in the frequency of the b.f.o. shall result from counterclockwise rotation of this control from this latter marking.

3.16.28.2.1 The vernier frequency control shall provide a total variation of the frequency of the b.f.o. of not less than 2700 Hz nor more than 3300 Hz, over the complete range of control.

3.16.28.2.2 The range of the vernier frequency control shall embrace at least 170 degrees of angular rotation. The variation of frequency with respect to angular rotation of the control shall be substantially linear.

3.16.29 Frequency and gain stability. - The requirements for frequency and gain stability shall apply under conditions of adjustment of the radio receiving set for A-1, broad reception. The frequency stability figures shall apply to the variation in a 1000 Hz beat note with a constant r.f. input frequency. The frequency changes shall be noted with respect to the effects or results of the frequency variations of all oscillators on the beat note frequency. All figures refer to total variations, either instantaneous or gradual. The gain variation figures shall apply to overall receiver gain, including a.f. gain, but excluding changes due to frequency response characteristics of the a.f. amplifier. The gain variation figures shall apply to either instantaneous, gradual or transient changes. The following requirements shall be applicable at any frequency within the complete nominal frequency range of the equipment.

3.16.29.1 Sixty-minute warm-up period. - The frequency variation shall not exceed 0.05 percent from 14 to 30 kHz; 0.03 percent above 30 to 63 kHz; and 0.02 percent above 63 to 600 kHz.

3.16.29.1.1 The foregoing are predicated upon the receiver being in its closed condition (all panel operated fasteners secured) and with the receiver being preconditioned, with no power applied for a period of not less than 6 hours in a room where the temperature is maintained between 20° C and 30° C, prior to "turn-on" time. The power line voltage shall be maintained at 115 volts ±1 percent and the ambient temperature and humidity shall be maintained within 5 percent during the warm-up period. (See 4.5.310.)

3.16.29.2 Variation in primary power supply voltage and frequency. - The variation of power supply voltage and frequency over the limits and combinations of variations, as specified in paragraph titled "Supply line voltage and frequency" of MIL-E-16400, shall not cause a change in the resonant frequency of the receiver in excess of ±20 Hz or a change in receiver gain in excess of 1 db. The requirements of MIL-E-16400 with respect to transient voltage and frequency variations are not applicable.

3.16.29.3 Variation of ambient temperature. - The frequency variation shall not exceed 0.002 percent per degree centigrade or 2 Hz per degree centigrade, whichever is least exacting, when the equipment is subject to the temperature cycling tests of MIL-E-16400 for class 4 temperature. The gain variation shall not exceed 12 db during the same temperature cycling.

3.16.29.4 Variation of relative humidity. - The variation in frequency shall not exceed 150 Hz and the variation in gain shall not exceed 3 db during the humidity cycle following the temperature cycling of the humidity test specified in MIL-E-16400.

3.16.29.5 Control variations. - Frequency variations resulting from any variations of any of the following controls throughout its operating range, shall not exceed ±5 Hz.

(a) Output
(b) Gain
(c) Antenna compensator (compt)
(d) I.f. selectivity
(e) A.f. selectivity
(f) Reset of band change switch

3.16.30 Regeneration. - The interstage gain shall be so proportioned that there shall be no evidence of unintended oscillations or instability under any mode of reception, with the primary line voltage 10 per cent

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higher than the nominal applied voltage and with the receiver resonant to the r.f. input frequencies, for which the receiver offers the highest gain, in their respective frequency bands.

3.16.30.1 Under the foregoing, the variation in band widths of overall selectivity, as determined at the 60 db points of attenuation, shall not exceed 10 percent between the band widths obtained for adjustment of the r.f. gain control for standard gain and maximum gain.

3.16.31 Oscillator radiation. - Signals originating from any oscillator within the equipment shall not exceed a level of 400 microwatts, when measured at the antenna connector, in accordance with the procedure specified in MIL-I-16910.

3.16.32 Time constant of recovery from surge inputs. - With the equipment adjusted for the reception of A-1 Broad reception, sharp pulses of c.w. signals, at resonance, with a maximum amplitude of 5 volts (to simulate "static" surges) shall not cause a reduction in sensitivity that will persist in excess of 0.5 second after cessation of the input surge. The recovery time shall be that time required for 63 per cent of nominal sensitivity. The 5 volts shall be applied only to the high impedance input.

3.16.33 Inherent automatic gain control (a.g.c.) action. - The radio receiving set shall be capable of receiving A-1, A-2, and F-1 signals, as pertinent, with any r.f. carrier input up to 100,000 microvolts, applied to the high impedance antenna connection, without audio amplifier or detector blocking. The inherent a.g.c. action shall become effective in less than 0.20 second after application of an r.f. carrier input of 100,000 microvolts and, unless substantially instantaneous in action, shall have a restoration period of not more than 0.20 second. Time of operation or recovery shall be determined on the basis of 63 percent of the steady amplitude of the final effect. Start of time shall be based on the moment of application or removal of the test signal.

3.16.34 Thermal cutout. - The thermal cutout, associated with the low impedance antenna circuit, shall be hermetically sealed. It shall operate in such a manner that it will remain closed for ambient temperatures of 60° C and below with 1 ma or less, current flow. At 25° C ambient, it shall open within 30 seconds with 2.0 ampere current flow through the cutout.

3.16.35 Low pass antenna filter. - The low pass filter, associated with the low impedance input circuit, shall comply with the following requirements:

(a) Attenuation over pass band from 14 kHz to 600 kHz - 2 db maximum
(b) Attenuation from 1 megaHertz (MHz) to 5 MHz - not less than 30 db
(c) Attenuation from 5 MHz to 32 MHz - not less than 50 db

3.16.36 Power supply r.f. filters. - The r.f. filters, provided between the primary power input connector and the power supply transformer, shall afford not less than 100 db attenuation of any signal which may enter the radio set through circuits or parts other than at the antenna input connector. The specified minimum attenuation shall be with respect to standard A-1 sensitivity, as measured through the antenna input connector. It shall include the attenuation offered to the undesired signals by the power supply components.

3.16.37 Audio output low pass filter. - The a.f. output low pass filter shall be of such design as to attenuate r.f. signals entering the receiver via the a.f. output connectors, by not less than 100 db, over the frequency range from 100 kHz to 50 MHz.

3.16.38 Shielding integrity. - The radio receiving set shall be effectively shielded so that when the antenna input terminals are isolated with a well shielded dummy antenna, it shall be capable of operation in an r.f. field of 2 volts per meter, at any frequency throughout its nominal frequency range, when conditioned for the measurement of A-1 sharp sensitivity, at the same frequency, without the audio output exceeding 6 milliwatts in a 600-ohm load.

3.16.39 Inclination vibration and shock. - The complete equipment shall be so designed and constructed as to comply with the following requirements:

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(a) Inclination - As specified in MIL-E-16400
(b) Vibration - As specified for type I in MIL-E-16400
(c) Shock - In accordance with MIL-S-901 for grade A, class I, type A, deck-mounted, lightweight equipment tests

3.16.39.1 The variation in frequency for any mode of reception shall not exceed 0.02 percent of the resonant frequency when the equipment is subjected to inclination and vibration - with each of these influences acting independently.

3.16.39.2 Intermodulation products, under conditions of inclination and vibration (each acting separately), shall not exceed an amplitude greater than 30 decibels below the amplitude of the a.f. output signal for any mode of reception.

3.16.40 Oscillator activity. - In the design and adjustment of oscillators (including calibration oscillator) to provide suitable values of excitation, the oscillators shall be operated as actively as is consistent with the requirements of this specification.

3.16.41 Radio interference and undesired radiation. - Radio interference and undesired radiation shall conform to the requirements of MIL-E-16400 and class IA of MIL-STD-461.

3.16.42 Magnetic characteristics. - The radio receiving set shall comply with the requirements of MIL-E-16400 for minimum magnetic signature to the extent that such compliance is practical and compatible with the Government-furnished equipment, without major redesign.

3.16.43 Workmanship. - Workmanship shall conform with the requirements of MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective inspection and quality assurance system in accordance with MIL-Q-9858 and to the extent acceptable to the Government, covering the supplies under contract. A current written description of the system shall be submitted to the cognizant Government inspector for system approval prior to first article inspection. Any changes to the approved quality assurance system, which might affect the degree of assurance required by this specification or other applicable documents, must be submitted to the cognizant Government inspector for written approval prior to use.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of product to prescribed criteria has been demonstrated.

4.2 General inspection. - The general inspection of the equipment shall be composed of all inspections of of MIL-E-16400, except as otherwise specified herein.

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4.3 First article inspection. - First article inspection shall be conducted in accordance with paragraph titled "Preproduction inspection" of MIL-E-16400, except that inspection for water cooling and reliability will not be required. In addition, the following inspections shall be conducted to determine compliance with the requirements of this specification. Measurements shall be made at the frequencies indicated for each inspection. The test procedures of 4.5 through 4.5.3.10 are applicable, as pertinent, to the specified inspections.

4.3.1 Frequency band coverage. - Measurements shall be made to determine conformance with 3.6.1 and table I.

4.3.2 Frequency overlap. - Measurements shall be made to determine conformance with 3.16.1 through 3.16.1.2 and tables II and III.

4.3.3 Accuracy of digital main tuning indicator calibration. - Adjust BFO to 201.000 kHz in "A-1 Broad" reception mode and tune receiver for resonance with the applied r.f. signal. Adjust internal calibrator to closest point to the frequency being tested prior to each measurement. Measurements shall be made for five r.f. input frequencies, in geometric progression for each frequency band, starting and ending at the nominal end frequencies for each frequency band under measurement. Measurements shall be made for both clockwise and counterclockwise rotation of the main tuning control, in each instance, in the determination of conformity with 3.16.2 and table IV and 3.16.4.

4.3.4 Linearity of tuning variation. - Measurements to determine compliance with 3.16.3 shall be conducted for each 10 per cent of rotation of the ganged tuning capacitors. Measurements shall be applicable for band V only with radio set conditioned as for "A-1 Broad" reception.

4.3.5 Sensitivity. - Measurements shall be made at the nominal end frequencies and at the geometric mean frequency for each frequency band and for all modes of reception for both low and high antenna inputs (see 3.16.5 and table V). In addition, a minimum of 3 sensitivity measurements shall be made in each frequency band with the equipment powered from a 400 Hz primary power source.

4.3.6 Gain variation. - Measurements shall be made at the frequencies of maximum and minimum gain in each frequency band and for each mode of reception (see 3.16.6).

4.3.7 Reserve gain. - Measurements shall be made at the frequency of lowest gain in each frequency band and for each mode of reception (see 3.16.7).

4.3.8 Preselection selectivity. - Measurements shall be made at the nominal end frequencies of each frequency band for both low and high antenna impedance operation. (See 3.16.8 and table VI.) Measurements shall be made at the cathode of the first conversion (mixer) electron tube, with the local oscillator disabled.

4.3.9 Overall r.f. and i.f. selectivity. - Measurements shall be made at the nominal end frequencies of each frequency band, for both high and low antenna impedance operation (see 3.16.9 and table VII).

4.3.10 Final i.f. selectivity. - Measurements shall be made for both BROAD and SHARP positions of the i.f. selectivity control, with the GAIN control adjusted as for sensitivity measurement and with the gain reduced 60 db below that gain adjustment (see 3.16.10 through 3.16.10.3) for an applied i.f. frequency of 200 kHz (nominal).

4.3.11 Overload (desensitization) selectivity. - Measurements shall be made at the nominal low and high frequency ends of each frequency band with low impedance antenna input operation (see 3.16.11 and table VIII and 4.5.3.1).

4.3.12 Cross modulation selectivity. - Measurements shall be made at 250 and 600 kHz, high impedance antenna input and with the receiver conditioned for A-2 reception. (See 3.16.12 and table IX and 4.5.3.2.)

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4.3.13 I.f. rejection. - Measurements shall be made at the nominal high and low frequency end of each frequency band and for both high and low impedance antenna operation (see 3.16.13).

4.3.14 Image rejection. - Measurements shall be made at the same frequencies and under the same conditions as for 4.3.13 (see 3.16.14).

4.3.15 Other spurious responses. - With the receiver conditioned for "A-1 Broad", low impedance antenna input, measure as follows:

(a) With no r.f. signal input - explore each frequency band to determine compliance with 3.16.15 and 3.16.15.1.

(b) Measure spurious responses for the nominal end frequencies and the geometric mean frequency of each frequency band resulting from off resonant frequencies of 100-microvolt input. For each resonant adjustment of the receiver the range or resonant frequencies shall embrace the full frequency spectrum from 14 to 600 kHz. (See 3.16.15 and 3.16.15.2.)

4.3.16 Audio output voltage regulation. - Audio output variations, with change of load impedance, shall be made at the line output and head phone terminals (see 3.16.16).

4.3.17 Resonant overload output. - Measurements shall be made at the nominal high frequency end of each frequency band with the receiver conditioned for A-1 Sharp, A-1 Broad and F-1 reception and for low and high impedance antenna input (see 3.16.17 through 3.16.17.3 and 4.5.3.3).

4.3.18 Maximum undistorted output. - Measurement shall be made at the nominal high frequency end of frequency bands IV and V with the receiver conditioned for A-2 mode of reception, and for low and high impedance antenna input (see 3.16.18 through 3.16.18.2 and 4.5.3.4).

4.3.19 Audio amplifier frequency response. - The audio amplifier shall be measured from the final detector to the audio line and head phone output terminals (600-ohm load) over the range of audio frequencies from 50 to 10,000 Hz, for both the broad and sharp positions of the a.f. selectivity control. (See 3.16.19 through 3.16.19.4 and 4.5.3.5.)

4.3.20 C.w. detector distortion. - Measurements shall be made with receiver conditioned for "A-1 Sharp" mode of reception and for an applied r.f. input of 500 microvolts at 500 kHz (see 3.16.20 and 4.5.3.6).

4.3.21 Output signal-to-hum ratio. - Measurements shall be made at the nominal end frequencies of each frequency band, with the receiver conditioned for "A-1 Broad" mode of reception, for both 60 and 400 Hz power line frequencies (see 3.16.21 and 3.16.21.1).

4.3.22 Manual r.f. gain control characteristics. - The range of the manual r.f. gain control shall be measured at the nominal high and low ends of each frequency band and for each mode of reception. Linearity shall be measured at 500 kHz with the receiver conditioned for "A-1 Broad" reception and low impedance antenna input (see 3.16.22 through 3.16.22.2 and 4.5.3.7).

4.3.23 Output limiter characteristics. - The output limiter characteristics shall be determined with the receiver conditioned for "A-1 Broad" mode of reception, manual r.f. gain control adjusted for standard gain, and an applied r.f. signal of 500 kHz for high and low impedance antenna operation. Measure for compliance with 3.16.23 through 3.16.23.2 for limiting characteristic, distortion and time constant for limiting (see 4.5.3.8).

4.3.24 Range and linearity of output limiter level control. - Measure at same frequency and same mode of operation as for 4.3.23 except measure for line output loads of 600 and 300 ohms (see 3.16.24 and 3.16.24.1).

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4.3.25 Noise limiter. - Measure at nominal low and high frequency ends of each frequency band to determine compliance with 3.16.25, with receiver conditioned for high impedance antenna and A-2 mode reception.

4.3.26 Range and linearity of head telephone level control. - Measure for an applied r.f. signal of 500 kHz with receiver conditioned for high impedance antenna input, and for A-1 Broad and A-2 modes of reception (see 3.16.26).

4.3.27 Range and linearity of a.f. output level control. - The range and linearity of the a.f. output level control shall be measured with the output limiter inoperative; in the A-2 and FSK modes of operation; an r.f. input signal 500 kHz and for resistive line output loads of 600 and 300 ohms (see 3.16.27).

4.3.28 Beat frequency oscillator and frequency vernier. - With the receiver conditioned for low impedance antenna operation and an applied r.f. signal of 500 kHz (carrier), measure for compliance with 3.16.28 and 3.16.28.1 for A-1 and F-1 modes of reception, as applicable. With the receiver conditioned for A-1 reception determine compliance with 3.16.28.2 through 3.16.28.2.2 for range and linearity of the b.f.o. frequency vernier. Data shall be obtained for every 2½ degrees of control dial readings, with the vernier control rotated counterclockwise and clockwise, each with respect to the zero dial setting.

4.3.29 Frequency and gain variations. - Measurements to determine compliance with 3.16.29 through 3.16.29.5 shall be conducted at the nominal low and high frequency ends of each frequency band, with the receiver conditioned for low impedance antenna connection, and A-1 Broad mode of reception (see 4.5.3.10).

4.3.29.1 Measurement of sensitivity and overall selectivity shall be made before the start and immediately after completion of the temperature test, as required by MIL-E-16400.

4.3.29.2 Measurement of sensitivity and overall selectivity shall be made before the start and immediately after completion of each humidity cycle.

4.3.30 Regeneration. - Measurements shall be conducted with the receiver conditioned for A-1 Sharp mode of reception, low impedance antenna input operation, power supply line voltage 10 per cent higher than the nominal applied voltage, and with the receiver resonanted to the r.f. frequency, in each frequency band, where the receiver offers the highest gain (see 3.16.30).

4.3.31 Oscillator radiation. - Explore each frequency band for evidence of noncompliance with 3.16.31 for both low and high impedance antenna input operation.

4.3.32 Time constant of recovery from surge inputs. - Measure to determine compliance with 3.16.32 for an applied r.f. signal of 500 kHz at low impedance antenna input connection.

4.3.33 Inherent a.g.c. action. - With an applied r.f. signal input of 100,000 microvolts, at the nominal low and high frequency ends of each frequency band, low impedance antenna input, measure time of operation and recovery (after removal of the applied signal) for each of the specified modes of reception (see 3.16.33).

4.3.34 Antenna thermal cut-out. - Conformance with 3.16.34 shall be demonstrated in a complete equipment. The thermal cut-out shall function and the associated equipment shall not have been damaged.

4.3.35 Low pass antenna filter. - The frequency response of the low pass antenna filter shall be measured over the frequency range of 14 kHz to 32 mHz to determine conformance with 3.16.35. Data shall be recorded at input frequencies sufficient to define characteristic on a graphical plot (see 4.5.3.9).

4.3.36 Power supply r.f. filters. - With the receiver conditioned for A-1 Broad mode of reception and tuned for a resonant frequency of 600 kHz, apply an r.f. input signal of 10 volts at the power supply input connector. Measure attenuated signals at the outputs of the power supply modules. Apply r.f. inputs at 14, 25, 50, 100, 500 kHz and 1, 2, 4, 8, 16, and 32 mHz.

4.3.37 Audio output low pass filter. - Determine compliance with 3.16.37 for r.f. frequencies of 100, 200, 300, 500 kHz and 1, 2, 4, 8, 16, 32, and 50 mHz.

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4.3.38 Shielding integrity. - With the use of shielded high and low impedance simulated antennas and with the receiver resonant at 600 kHz, and maximum a.f. output in the A-1 Sharp condition, measure for compliance with 3.16.38.

4.3.39 Inclination vibration and shock. - Conduct tests to determine compliance with 3.16.39 through 3.16.39.2. Measurements for frequency variation and intermodulation products shall be made at the nominal low and high frequency ends of each frequency band, with the receiver conditioned, for low impedance antenna and A-1 Broad mode of reception.

4.3.40 Accelerated life test. - Accelerated life tests shall be conducted as specified in MIL-E-16400. Measurements of sensitivity with the receiver conditioned for low impedance antenna operation, shall be made for each mode of reception, concurrently with the recording of "HOT-SPOT" temperatures, as required in MIL-E-16400.

4.3.41 Oscillator activity. - Measure oscillator injection voltages at nominal low and high frequency ends of each frequency band, for applied power supply voltages of 115 volts and at 10 percent below this voltage (see 3.16.40).

4.4 Quality conformance inspection. -

4.4.1 Production inspection. - The following production inspections shall be conducted, in addition to those specified in MIL-E-16400 on each production equipment offered for Government acceptance, to determine compliance with this specification. These inspections are not applicable to equipments selected for production control inspection (see 4.4.2). The requirements of 4.5 through 4.5.3.10 are applicable, as pertinent to the specified inspections.

(a) Frequency overlap - Same as for 4.3.2.
(b) Accuracy of digital main tuning indicator calibration - Same as for 4.3.3.
(c) Sensitivity - Same as for 4.3.5.
(d) Gain variation - Same as for 4.3.6.
(e) Final i.f. selectivity - Same as for 4.3.10.
(f) I.f. rejection - Same as for 4.3.13, except for low impedance antenna input only.
(g) Image rejection - Same as for 4.3.14, except for low impedance antenna input only.
(h) Audio output regulation - Same as for 4.3.16.
(i) Resonant overload output - Same as for 4.3.17 except for 500 kHz r.f. input with receiver conditioned for A-1 Sharp and low impedance antenna input.
(j) Maximum undistorted output - Same as for 4.3.18, except for r.f. input of 500 kHz and low impedance antenna input only.
(k) C. W. detector distortion - Same as for 4.3.20.
(l) Output signal to hum ratio - Same as for 4.3.21.
(m) Manual r.f. gain control characteristics - Same as 4.3.22, except range and linearity shall both be determined for an applied r.f. signal of 500 kHz.
(n) Output limiter characteristics - Same as for 4.3.23, except for low impedance antenna input only.
(o) Range and linearity of output limiter level control - Same as for 4.3.24, except for 600-ohm line output load only.
(p) Range and linearity of a.f. output level control - Same as for 4.3.27, except F-1 mode of reception and 600-ohm line output load only.
(q) Beat frequency oscillator and frequency vernier - Same as for 4.3.28, except measure to determine compliance with 3.16.28 through 3.16.28.2.1 only, for A-1 Sharp and F-1 modes of reception, as pertinent. Measurement of linearity of frequency vernier is not required.
(r) Vibration - Each equipment shall be subjected to a 30-minute vibration test in each of three perpendicular planes. The amplitude of vibration shall be 0.010 ±0.002 inch at a frequency of 33 Hz. Measure for compliance with 3.16.39.2 for intermodulation products.

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(s) Burn in - Each equipment shall be operated for a continuous period of 72 hours, with full power applied, prior to start of production inspection. All deficiencies which develop during this burn-in test shall be appropriately corrected, to the satisfaction of the cognizant Government inspector. Like corrective measures shall be immediately included in all subsequent production units, yet to be fabricated or undelivered, at no cost to the Government.

4.4.2 Production.control inspection. - Production control inspection shall be conducted on one equipment, selected at random by the cognizant Government inspector from each lot of 50 production equipments or each months production lot, whichever is the lesser. Inspection shall include all inspections specified under production control inspection of MIL-E-16400, except for the specified inspections for water cooling and reliability. In lieu of the reliability requirements specified under production control inspection of MIL-E-16400, the reliability requirements of this specification shall govern. In addition, the following inspections shall be conducted to determine compliance with the specification. The requirements of 4.5 through 4.5.3.10 are applicable, as pertinent, to the specified inspections.

(a) Frequency overlap - Same as for 4.3.2.
(b) Accuracy of digital main tuning indicator - Same as for 4.3.3.
(c) Linearity of tuning variation - Same as for 4.3.4.
(d) Sensitivity - Same as for 4.3.5.
(e) Gain variation - Same as for 4.3.6.
(f) Reserve gain - Same as for 4.3.7.
(g) Preselection selectivity - Same as for 4.3.8, except at the nominal high frequency end of each frequency band, and for low impedance antenna input only.
(h) Overall r.f. selectivity - Same as for 4.3.9, except at the nominal high frequency end of each frequency band, and for low impedance antenna input only.
(i) Final i.f. selectivity - Same as for 4.3.10, except for low impedance antenna input only.
(j) Overload (desensitization) selectivity - Same as for 4.3.11 except at nominal high frequency end of each frequency band.
(k) Cross modulation selectivity - Same as for 4.3.12, except for r.f. input frequency of 500 kHz.
(l) Lf. rejection - Same as for 4.3.13.
(m) Image rejection - Same as for 4.3.14.
(n) Other spurious responses - Same as for 4.3.15.
(o) Audio output voltage regulation - Same as for 4.3.16.
(p) Resonant overload output - Same as for 4.3.17.
(q) Maximum undistorted output - Same as for 4.3.18.
(r) Audio amplifier response - Same as for 4.3.19.
(s) C.W. detector distortion - Same as for 4.3.20.
(t) Output signal-to-hum ratio - Same as 4.3.21.
(u) Manual r.f. gain control characteristics - Same as for 4.3.22, except at the nominal high frequency end of eacy frequency band only.
(v) Output limiter characteristics - Same as for 4.3.23.
(w) Range and linearity of output limiter level control - Same as for 4.3.24.
(x) Noise limiter - Same as for 4.3.25, except for an r.f. signal of 500 kHz.
(y) Range and linearity of head telephone level control - Same as for 4.3.26 except for A-1 Broad mode only.
(z) Range and linearity of a.f. output level control - Same as for 4.3.27 except for A-1 Broad mode only.
(aa) Beat frequency oscillator and frequency vernier - Same as for 4.3.28.
(bb) Regeneration - Same as for 4.3.30.
(cc) Oscillator radiation - Same as for 4.3.31.
(dd) Antenna thermal cutout - Same as for 4.3.34.
(ee) Antenna low pass filter - Same as for 4.3.35.
(ff) Power supply r.f. filters - Same as for 4.3.36 except measure at test frequency to which the filter offers the least attenuation.
(gg) Audio output low pass filter - Same as 4.3.37.

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(hh) Burn-in - The equipment shall be operated for a continuous period of 72 hours, with full power applied, prior to start of production control inspection. All deficiencies which devleop during this burn-in test shall be appropriately corrected, to the satisfaction of the cognizant Government inspector. Like corrective measure shall be immediately included in all production units represented by this lot and all subsequent production units yet to be fabricated or undelivered, at no cost to the Government.

4.4.3 Environmental inspection. - One equipment out of the first 25 production units and one equipment out of the subsequent 100 production equipments, or nearest fraction thereof, shall be subjected to all of the environmental inspection tests specified in MIL-E-16400.

4.4.4 Reliability demonstration for production equipment. -

4.4.4.1 Reliability demonstration test plans. - Sixty days prior to setting up the test equipment, the contractor shall submit to the command or agency concerned a reliability demonstration test plan which conforms to the Detail Requirements section of MIL-STD-781 and the requirements of this specification.

4.4.4.2 Reliability qualification tests. - Qualification testing shall be performed on at least the first five production units. Testing shall be performed at test level C with test plan V. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan V. "On-off" cycling, as specified in MIL-STD-781, is not required.

4.4.4.3 Reliability sampling tests. - Reliability sampling tests shall be conducted throughout production to insure continued production of acceptable equipment. Sampling testing shall be at test level C with test plan VI of MIL-STD-781. At least three units shall be randomly selected from each lot to be delivered for reliability sampling tests. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan VI. "On-off" cycling, as specified in MIL-STD-781, is not required.

4.4.4.4 Failure. - Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification.

4.4.4.5 Rejected equipment. - When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be rejected. In the event of rejection, the lot of equipments shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. No equipment shall be selected more than once for reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer design, corrective action shall be performed on all equipments delivered under the contract. If failure is due to equipment manufacturer workmanship or part manufacturer workmanship, the entire inspection lot shall be inspected for the discovered fault and reworked as necessary.

4.4.4.6 Disposition of tested equipment. - Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of section 4.4 to assure that it meets the requirements of this specification. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the contract for acceptance.

4.4.4.7 Test parameters. - As a minimum, the parameters to be monitored during reliability testing shall be as follows:

Test	Requirement paragraph	Test paragraph
Accuracy of main tuning calibration	3.16.2	4.3.3
Sensitivity	3.16.5	4.3.5
Resonant overload output	3.16.17	4.3.17

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4.5 Test procedures and definitions. - The following test procedures and definitions are applicable to the requirements of this specification for first article, production, production control, and environmental inspections, as well as reliability observations in the determinations of compliance with the performance requirements specified herein. The test procedures included herein do not relieve the contractor of the responsibility for the preparation of complete test procedures to be submitted for Government approval, in accordance with MIL-E-16400.

4.5.1 Standard test conditions. - Unless otherwise specified herein, all tests conducted to determine conformity with the performance requirements of this specification, shall be conducted under the following standard conditions:

(a) Power source - 115 volts, 60 Hz, single phase a.c. except for reliability tests, where the power source shall be 115 volts ±10 percent at 400 Hz ±5 percent.
(b) R.f. input - Through low and high impedance antenna simulators, as pertinent to specified individual tests.
(c) A.f. output - 600 ohms, noninductive resistive load across one line output connector.
(d) R.f. selectivity - As specified.
(e) A.f. selectivity - As specified.
(f) BFO and frequency vernier - As specified.
(g) A.f. output level control and head telephone level control - As specified.
(h) R.f. gain - As specified.
(i) Antenna compensator - Adjust to optimum setting but once for each frequency band.
(j) Noise limiter - Off, except as otherwise specified.
(k) Ambient temperature - Between 20° and 30° centigrade.
(l) Relative humidity - Below 50 percent.
(m) Electron tubes - Within limits specified in MIL-STD-200.

4.5.2 Definitions. -

4.5.2.1 Dummy 50-ohm antenna simulator (low impedance). - A dummy antenna simulator, for low impedance antenna input connection, shall, in combination with the signal generator impedance, be equivalent to 50 ohms. A signal generator, having an internal impedance of 50 ohms, may be connected directly, without additional terminating resistors. The signal generator output indication shall be connected to read open circuit voltage; for example, the output indication shall be multiplied by 2 if the signal generator has an internal impedance of 50 ohms and is calibrated to read actual voltage across a 50-ohm termination.

4.5.2.2 High impedance dummy antenna simulator. - The high impedance dummy antenna simulator shall consist of an inductance of 20 microhenries in shunt with a series combination of a 400-picofarad capacitor and a 400-ohm resistor, the entire combination being in series with a 200-picofarad capacitor.

4.5.2.3 Standard output signal plus noise-to-noise ratio. - Standard output signal plus noise-to-noise ratio shall be 10 decibels (db) for A-2 reception and 20 db for A-1 and F-1 signals. These ratios shall be determined in terms of root mean square (rms) values.

4.5.2.3.1 For A-2 mode of reception, the output signal plus noise-to-noise ratio shall be the ratio between a standard audio output signal, obtained under conditions of measurement for MCW sensitivity, and the output due to noise when the modulation is removed from the r.f. carrier.

4.5.2.3.2 For A-1 and F-1 mode of reception, the output signal plus noise-to-noise ratio shall be the ratio between a standard audio output, obtained under conditions of measurement for A-1 or F-1 sensitivity, and the output due to noise with the carrier removed.

4.5.2.4 Standard audio output. - Standard audio output shall be 6 milliwatts of power dissipated into a 600-ohm, noninductive resistance across the a.f. line output terminals.

4.5.2.5 Standard gain. - Standard gain shall be that gain which produces standard output signal plus noise-to-noise ratio with standard audio output.

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4.5.2.6 Standard modulation frequency. - Standard modulation frequency shall be 1000 Hz.

4.5.2.7 Standard beat note frequency. - Standard beat note frequency for c.w. reception shall be 1000 Hz.

4.5.2.8.1 Standard A-2 sensitivity. - Standard A-2 sensitivity shall be the number of microvolts of pure c.w., r.f. energy, modulated 30 percent at standard modulation frequency (applied to the antenna input through a simulated dummy antenna) required to produce a standard audio output under conditions of standard output signal plus noise-to-noise ratio, as defined in 4.5.2.3.1 and as determined by adjustment of the r.f. and i.f. gain control with the a.f. level control set at maximum output level.

4.5.2.8.2 Standard A-1 or F-1 sensitivity. - Standard c.w. sensitivity shall be the number of microvolts of pure c.w., r.f. energy (applied to the antenna input through a simulated dummy antenna) required to produce a standard audio output, under conditions of standard output signal plus noise-to-noise ratio, as defined in 4.5.2.3.2 and as determined by adjustment of the r.f. and i.f. gain control with a.f. output level set at maximum output level.

4.5.2.9 Selectivity. - Selectivity of a radio receiver is that characteristic which determines the extent to which it is capable of differentiating between the desired resonant signal and r.f. signals of other frequencies.

4.5.2.9.1 Weak signal selectivity. - Weak signal selectivity shall be the degree to which the radio receiving set is capable of differentiating between the desired signal and relatively weak interfering signals of frequencies relatively close to the desired signal frequency. It shall be expressed in terms of the attenuation, in db, which the radio receiving set introduces to frequencies removed from resonance, with respect to the resonant sensitivity, using standard audio output as a common reference level. This definition is applicable to preselection, overall r.f. and i.f., and i.f. selectivities.

4.5.2.9.2 Overload (desensitization) selectivity. - Overload (desensitization) selectivity is a measure of the degree to which a receiver is capable of preventing an undesired off-resonant signal from changing the gain or increasing the noise output of the receiver to a degree that is detrimental to the desired signal at the resonant frequency. This dual signal characteristic is measured with a constant desired signal of a known intensity, centered at the resonant frequency, and with unmodulated, undesired signals of higher intensities present, in turn, at various displacements from the resonant frequency. A change of 3 db in receiver desired signal output, or output noise (whichever appears first) is used as the reference output condition for determining the locus of the points of the overload selectivity curve. The desired signal output is standard output, for A-1 signal-to-noise ratio, with the undesired signals absent.

4.5.2.9.3 Cross-modulation selectivity. - The cross-modulation selectivity of a receiver is a measure of its ability to prevent the transfer of modulation from an undesired, modulated, off-resonant signal to a desired signal, appearing at the resonant frequency. This two-signal characteristic may be measured with both signals modulated equally. The off-resonant, undesired signal is varied in carrier frequency and intensity so as to cause the appearance of cross-modulation output which is 10 db below the standard output produced from the desired signal. This degree of interference, which is cross-modulation (not "cross talk"), is the reference output condition for determining the locus of the cross-modulation selectivity curve. The desired signal output is initially set at standard output level for A-2 signal-to-noise ratio with the undesired signal absent, and its modulation removed during the measurements.

4.5.2.9.4 Intermodulation products. - Intermodulation products of a receiver is the measure of its ability to prevent excessive interference caused by intermodulation products, which are generated in the receiver through the interaction of off-resonant input signals. In order to produce such interference the undesired signals shall have a frequency spacing relationship that results in the generation of a component at or near the desired signal frequency, through a combination of interfering signal frequency multiplication and inter-

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signal heterodyning. At least two undesired signals, having such a particular relationship, shall be simultaneously present for this type of interference to appear, but the desired signal need not be present. The so-called 2A-B forms of intermodulation produce the greatest effect. The selectivity characteristics for the 2A-B forms of intermodulation is measured using two unmodulated input signals of equal intensity, spaced f and 2 f, in frequency, on the same side to produce a particular equivalent c.w. signal input at the resonant frequency, through intermodulation. The standard reference equivalent signal input shall be 10 db below standard A-1 sensitivity signal-to-noise ratio and standard output. The intermodulation selectivity curve is the locus of the points defining the frequency separation of f for the various equal, undesired signal levels that produce the reference value of equivalent input.

4.5.2.10 Spurious responses. -

4.5.2.10.1 I.f. rejection. - I.f. rejection shall be the degree to which the radio receiving set is capable of rejecting signals having the frequency(s) of the i.f. amplifier(s). I.f. rejection shall be expressed in terms of the attenuation, in db., the radio receiving set introduces to i.f. frequencies, with respect to the resonant sensitivity, using standard output as the reference level, with the receiver adjusted for A-1 sensitivity.

4.5.2.10.2 Image rejection. - Image rejection shall be that degree to which the radio receiving set is capable of rejecting signals off resonance which, in combination with the fundamental or any harmonic of the conversion oscillator(s), produce frequencies which are amplified by the i.f. amplifier(s), and result in spurious responses. It shall be expressed in terms of the attenuation, in db, which the radio receiving set introduces to the image frequencies, with respect to the resonant A-1 sensitivity, using standard audio output as a common reference level. Image rejection shall be relative to the resonant frequency rather than the image frequency.

4.5.2.11 Resonant overload. - Resonant overload with a.f. output level at maximum, shall be that a.f. output which occurs when the overall gain of the receiver, as adjusted for A-1 sensitivity and standard output, has decreased by 3 db with increase in r.f. carrier input level, from zero input.

4.5.2.11.1 Resonant overload blocking shall be that point where the gain of the receiver decreases at a rate greater than 1 db per db of increase in resonant carrier input.

4.5.2.12 Maximum undistorted output. - Maximum undistorted output shall be the output power level with an input carrier, modulated 30 percent, at standard modulation frequency, at which the total output harmonic content is 7 percent, in voltage, with the radio receiving set adjusted as for measurement of standard A-2 sensitivity.

4.5.2.12.1 The total harmonic content shall be that measured by the elimination of the voltage of the fundamental modulation frequency, the resultant harmonic voltage components being measured by means of a wave analyzer, or equivalent, at the fundamental frequency.

4.5.2.13 Threshold of operation of output limiter. - The threshold of operation of the output limiter shall be the r.f. input level at which the overall gain of the receiver, as adjusted for A-1 or F-1 sensitivities, is decreased by 1 db.

4.5.3 Test procedures. -

4.5.3.1 Overload (desensitization) selectivity. - An unmodulated desired r.f. carrier shall be applied to the antenna input connector, through specified antenna simulators, and adjusted for standard A-1 sensitivity, with the receiver conditioned for A-1 mode of reception. An unmodulated interfering signal, adjusted to 60 db, 80 db, 100 db, and 120 db above the desired resonant signal level, shall be applied to the same antenna simulator, and the frequencies shall be measured, above and below resonance, at which the interfering signal causes a 3 db decrease in audio output. Sufficient isolation shall be provided between the applied desired signal and the interfering signal inputs to preclude cross-modulation occurring in the signal generator.

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4.5.3.2 Cross-modulation selectivity. - An r.f. input, representing a desired weak signal, modulated 30 percent at 1000 Hz, shall be applied through an antenna simulator to the antenna connector of the receiver. Sufficient isolation shall be provided between this desired signal source and the antenna simulator to preclude cross-modulation occuring in the signal generator. The level of this signal shall be adjusted for standard sensitivity, with the receiver conditioned for A-2 mode of reception. The modulation shall be removed from the desired signal. An interfering signal, modulated 90 percent at 1000 Hz, shall be applied to the same antenna simulator. The amplitude of the interfering signal shall be adjusted to 60 db, 80 db, and 100 db, above the resonant signal level, and its frequency measured, above and below resonance, at which the interfering signal modulation causes a 6 db increase in a.f. output.

4.5.3.3 Resonant overload output. - Measurements shall be made with receiver gain, initially adjusted and remained fixed for standard gain. Output level control at maximum. Output limiter and noise limiter at OFF. Vary applied r.f. carrier from zero up to 2.0 volts in incremental steps sufficient to clearly show resonant overload characteristics on a graphical plot of a.f. output versus r.f. input. The point at which resonant overload output occurs shall be indicated on the graphical plot. Repeat with the r.f. gain control adjusted for a constant gain level, 60 db below standard gain.

4.5.3.4 Maximum undistorted output. - Measurements shall be made of a.f. output for increases in modulated r.f. input from the input required for A-2 sensitivity up to 60 db above such input, with no adjustments being made of the receiver controls. The incremental changes in r.f. input shall be sufficient to clearly define the maximum undistorted output characteristics on a graphical plot of a.f. output versus r.f. input. Measure distortion for both the specified and "actual" maximum undistorted output. Indicate on a graphical plot the point of "actual" maximum undistorted output, i.e., the output where the distortion is 7 percent.

4.5.3.5 Audio amplifier frequency response. - Frequency response measurements shall include the audio circuits of the detector. The audio input voltage shall be applied to the detector output load through a resistive impedance representing the equivalent resistance of the detector, as an audio generator.

4.5.3.6 C.W. detector distortion. - With the receiver conditioned for "A-1 Sharp" and with the output level at maximum output, apply a 500-microvolt unmodulated signal at receiver resonance (500 kHz). Readjust r.f. gain for standard output. Adjust b.f.o. to 1000 Hz a.f. output. Readjust b.f.o. for 500 Hz a.f. beat note, and measure r.f. output level. Determine the db ratio between the output level for a 500 Hz beat note and the level for a 1000 Hz beat note.

4.5.3.7 Manual r.f. gain control range and linearity. - The range and linearity of the manual r.f. gain control shall be depicted on a graphical plot of attenuation, in db, versus angular rotation of control shaft, in degrees. The specification parameters for both range and linearity shall be included in such fashion to offer clear interpretation as to extent of compliance with these parameters. The graphical plot shall be addition to tabulated measurement data.

4.5.3.8 Output limiter characteristics. - Measurements shall be made for increases of r.f. carrier input from zero up to 500,000 microvolts, in incremental steps as to afford a well defined graphical plot (a.f. output versus r.f. input) of the output limiter characteristics. The graphical plot shall be marked with the point at which the threshold of operation becomes effective.

4.5.3.9 Output level control (output limiter) a.f. line output and head telephone output, range and linearity. - Present data, as graphical plots, in same manner, as specified in 4.5.3.7. These plots shall be in addition to tabulated measurement data.

4.5.3.10 R.F. input filter, power supply, r.f. filter and a.f. line filter. - Performance characteristics for the r.f. input filter, power supply r.f. filter and a.f. line filter shall be presented as graphical plots, in addition to corresponding tabulated measurement data.

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4.5.3.11 Warm-up frequency variation. - Measurement of warm-up frequency variations shall be made at intervals of not more than 2 minutes for the first 10 minutes, at intervals of not more than 5 minutes for the next 20 minutes, and at intervals of not more than 10 minutes for the final 30 minutes.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. - Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation, packaging, packing, or marking requirements, if other than as specified in 5.1.

6.2 First article. - Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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Navy - EC

(Project 5820-N450)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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